CSE276 - Artificial Intelligence Foundations | Section K3P2SOR | CA-1 Groups and Projects

Find your group by your name or registration number. The objectives are what your project will be assessed on; the advanced extension is optional. Use the official source to read the original problem statement (for Smart India Hackathon problems, search the PS ID on the linked page).

Group	Students	Project Title	Problem to Solve	Who Benefits	Objectives (what you will be assessed on)	Suggested Techniques	Syllabus Topics Used	Data to Use	Optional Advanced Extension	Inspired By (official source)
G1	Aditi (12510057) Deepika (12519640)	**Heavy Rainfall Early Warning and Urban Inundation Prediction**	Warnings are issued for rainfall but not for where water will actually stand, so cities evacuate the wrong streets or none at all.	City disaster cells, municipal drainage departments, residents	1. Build a warning system that forecasts heavy rainfall and converts it into a predicted inundation map with a graded alert. 2. Produce a rainfall-to-inundation model with alert lead time and zone-level accuracy against recorded events. 3. Report alert precision/recall on flood events, lead time, zone-level IoU. Baseline: rainfall-threshold alarm, mandatory. Deliver City map with predicted inundation zones and alert banner.	Rainfall forecasting from multi-source inputs; inundation classification per zone; alert thresholding	Unit VI time-series; Unit II multi-source feature engineering; Unit V rare-event metrics	IMD rainfall data, public urban drainage and flood-report datasets (Find and cite suitable public data yourselves.)	Couple drainage capacity so predicted standing water reflects where it can actually drain.	SMART INDIA HACKATHON 2026 PS ID: SIH26071 AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data. Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G2	Sidhima (12506381) Angad Singh (12505645)	**Building Footprint Mapping for Disaster Resilience**	Disaster planning in fast-growing African and Asian cities is blocked because nobody has an up-to-date map of what is built where.	City authorities, humanitarian mapping agencies	1. Build a segmentation system producing building footprints and a per-ward building density layer for risk planning. 2. Produce a footprint model with IoU and a produced density layer. 3. Report ioU, Dice, building count error per ward. Baseline: edge-detection segmentation. Deliver Web map with footprint overlay.	Instance and semantic segmentation on aerial imagery; polygonisation	Unit III-IV CNN segmentation; Unit II augmentation; Unit V segmentation metrics	Competition dataset - see the official page	Compare performance between planned and informal settlement areas.	DRIVENDATA Open Cities AI Challenge: Segmenting Buildings for Disaster Resilience Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/60/building-segmentation-disaster-resilience/ Dataset access: Competition dataset (platform account / rules acceptance)
G3	Ananya Sharma (12510850) Ishita (12507839)	**Roof Material Classification for Disaster Risk Assessment**	Storm vulnerability depends heavily on roof construction, and surveying it building by building is impossible at city scale.	Disaster risk agencies, insurers, municipal planners	1. Build a classifier that identifies roof material from aerial imagery and produces a neighbourhood vulnerability map. 2. Produce a roof material classifier with per-class recall and an aggregated vulnerability map. 3. Report per-class F1, macro-F1, map agreement with survey. Baseline: colour and texture features with SVM. Deliver Vulnerability map dashboard.	Image classification on aerial crops; class imbalance; geospatial aggregation	Unit III-IV CNN; Unit II preprocessing; Unit V multi-class metrics	Competition dataset - see the official page	Add uncertainty so low-confidence areas are flagged for survey.	DRIVENDATA Open AI Caribbean Challenge: Mapping Disaster Risk from Aerial Imagery Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/58/disaster-response-roof-type/ Dataset access: Competition dataset (platform account / rules acceptance)
G4	Abhay Negi (12514462) Reyansh Vijay (12517260)	**Hazard Red-Zone Identification and Relocation Planning**	Vulnerable habitations sit in known hazard zones and are relocated only after a disaster has already destroyed them.	Ministry of Home Affairs, district disaster authorities, at-risk habitations	1. Build a system that classifies terrain into hazard red zones, assesses how many people each safe zone can absorb, and proposes a relocation plan. 2. Produce a red-zone classifier plus a capacity-respecting relocation assignment for one district. 3. Report zone classification F1, assignment feasibility, population served within capacity. Baseline: distance-only nearest-safe-zone assignment. Deliver District map with red zones and proposed relocation links.	Multi-class hazard classification on geospatial features; capacity-constrained assignment	Unit II geospatial feature engineering; Unit III classification; Unit I optimisation and assignment	Bhuvan hazard layers, Census habitation data, OpenStreetMap (Find and cite suitable public data yourselves.)	Add cost and distance objectives so relocation is realistic, not just feasible.	SMART INDIA HACKATHON 2026 PS ID: SIH26191 Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations Organisation: Ministry of Home Affairs https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G5	Shradul Sharma (12508811) Rishabh Ranjan (12501098)	**Tropical Storm Intensity Estimation from Satellite Imagery**	Storm intensity drives evacuation decisions and is estimated by trained analysts reading imagery, which does not scale round the clock.	Meteorological agencies, disaster response, coastal authorities	1. Build an estimation system predicting wind speed from satellite imagery sequences with an uncertainty band. 2. Produce a wind speed regressor with RMSE against analyst estimates. 3. Report rMSE, MAE, error by intensity band. Baseline: single-frame regression. Deliver Storm tracking dashboard with intensity trace.	CNN regression on image sequences; temporal feature use; uncertainty estimation	Unit III-IV CNN; Unit VI time-series; Unit V regression evaluation	Competition dataset - see the official page	Use the image sequence rather than a single frame and report the gain.	DRIVENDATA Wind-dependent Variables: Predict Wind Speeds of Tropical Storms Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/72/predict-wind-speeds/ Dataset access: Competition dataset (platform account / rules acceptance)
G6	Amrit (12510248) Kirat Singh Walia (12506272)	**Geomagnetic Disturbance Forecasting for Grid Protection**	Geomagnetic storms induce currents that damage transformers and disrupt navigation, and operators want warning in hours not minutes.	Power grid operators, satellite and navigation services	1. Build a forecasting system predicting the geomagnetic disturbance index from solar wind measurements. 2. Produce a disturbance forecaster with RMSE against a persistence baseline at multiple lead times. 3. Report rMSE, MAE, storm-period skill. Baseline: persistence, mandatory. Deliver Space weather dashboard with alert bands.	Time-series regression from multivariate drivers; lag feature engineering	Unit VI time-series; Unit II feature engineering; Unit V forecast evaluation	Competition dataset - see the official page	Report skill separately for storm periods versus quiet periods.	DRIVENDATA MagNet: Model the Geomagnetic Field Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/73/noaa-magnetic-forecasting/ Dataset access: Competition dataset (platform account / rules acceptance)
G7	Radhika Rai (12501516) Ridhima Gagneja (12519738)	**Building Height Estimation from Single Overhead Images**	Height data is missing from most city models, which blocks flood modelling, solar planning and 3D mapping.	Urban planners, disaster modellers, mapping agencies	1. Build a system estimating per-pixel height and orientation from a single overhead image to build a coarse 3D city layer. 2. Produce a height estimation model with RMSE against reference elevation data. 3. Report height RMSE, orientation error, R-squared. Baseline: shadow-length geometric estimation. Deliver 3D visualisation of a reconstructed block.	Dense regression from imagery; geometry-aware learning	Unit III-IV CNN; Unit II preprocessing; Unit V regression metrics	Competition dataset - see the official page	Render a 3D city block from the predicted heights.	DRIVENDATA Overhead Geopose Challenge Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/78/overhead-geopose-challenge/ Dataset access: Competition dataset (platform account / rules acceptance)
G8	Yedu Krishna T M (12504891) Hardik Bhardwaj (12506048)	**Cloud Cover Detection for Usable Satellite Imagery**	Every downstream satellite analysis silently degrades when cloud is mistaken for ground, and cloud masks are the unglamorous fix.	Remote sensing teams, agricultural and disaster analytics providers	1. Build a cloud segmentation service that produces a usability mask and a per-scene usable-area percentage. 2. Produce a cloud mask model with IoU and a scene usability report. 3. Report ioU, Dice, false-ground rate. Baseline: brightness thresholding, mandatory. Deliver Scene viewer with cloud mask toggle.	Semantic segmentation on multi-band imagery; band feature engineering	Unit II multi-band preprocessing; Unit III-IV CNN; Unit V segmentation metrics	Competition dataset - see the official page	Add thin-cloud and shadow handling, where simple methods fail.	DRIVENDATA On Cloud N: Cloud Cover Detection Challenge Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/83/cloud-cover/ Dataset access: Competition dataset (platform account / rules acceptance)
G9	Navpreet Kaur (12505005) Kritika (12504372)	**Kelp Forest Extent Monitoring for Marine Conservation**	Kelp forests are collapsing in places and expanding in others, and nobody can monitor the change at coastline scale by eye.	Marine conservation agencies, coastal researchers	1. Build a monitoring system that segments kelp canopy from satellite imagery and reports extent change over time. 2. Produce a kelp segmentation model with Dice and an extent-change time series. 3. Report dice, IoU, extent error. Baseline: spectral index thresholding. Deliver Coastline map with extent trend chart.	Semantic segmentation on multispectral imagery; temporal change analysis	Unit II band preprocessing; Unit III-IV CNN; Unit V segmentation metrics	Competition dataset - see the official page	Separate genuine change from seasonal variation.	DRIVENDATA Kelp Wanted: Segmenting Kelp Forests Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/255/kelp-forest-segmentation/ Dataset access: Competition dataset (platform account / rules acceptance)
G10	Pratham Bharti (12519532) Rishika Garg (12518700)	**Camera-Trap Wildlife Census for Conservation Monitoring**	Conservation camera traps generate millions of images that volunteers label by hand, so population trends lag reality by years.	Wildlife conservation agencies, protected area managers, ecologists	1. Build a census system that identifies species in camera-trap images, discards empty frames, and produces population trend statistics. 2. Produce a species classifier with per-species recall and a produced trend series for one reserve. 3. Report macro-F1, per-species recall, empty-frame rejection rate. Baseline: motion-difference filter with a single-class model. Deliver Ranger dashboard with species counts and trend chart.	Image classification with a large species label set; empty-frame filtering; long-tailed class handling	Unit III-IV CNN and transfer learning; Unit II augmentation; Unit V multi-class and long-tail metrics	Competition dataset - see the official page	Handle rare species explicitly rather than letting the common ones dominate the metric.	DRIVENDATA Hakuna Ma-data: Identify Wildlife on the Serengeti with AI for Earth Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/59/camera-trap-serengeti/ Dataset access: Competition dataset (platform account / rules acceptance)

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G11	Ronak Khanna (12501517) Ansh Sood (12508720)	**Planetary Sample Spectrometry Analysis for Habitability Evidence**	Spectrometry data returned from Mars must be interpreted by a small number of specialists, which bottlenecks mission science.	Planetary science teams, mission operations	1. Build a classification system identifying chemical families present in a sample spectrum, with confidence. 2. Produce a multi-label spectral classifier with per-compound F1. 3. Report per-label F1, macro-F1, log loss. Baseline: peak-matching heuristic. Deliver Analysis tool showing spectrum with predicted families.	Multi-label classification on spectra; signal feature engineering	Unit II signal feature engineering; Unit III classification; Unit V multi-label metrics	Competition dataset - see the official page	Add abstention where the spectrum is too noisy to call.	DRIVENDATA Mars Spectrometry: Detect Evidence for Past Habitability Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/93/nasa-mars-spectrometry/ Dataset access: Competition dataset (platform account / rules acceptance)
G12	Nazim Shaikh (12514609) Vikas Kumar Verma (12512884)	**Research Literature Assistant for Mission Scientists**	Mission scientists cannot keep up with the literature relevant to their instrument, so prior findings are rediscovered.	Research agencies, mission science teams, librarians	1. Build a retrieval assistant that answers technical questions from a paper corpus and cites the passage used. 2. Produce a cited-answer assistant evaluated on questions with known answers. 3. Report answer correctness and citation correctness separately; refusal rate. Baseline: keyword search. Deliver Research assistant web interface.	Retrieval-augmented generation; semantic search; citation grounding	Unit V NLP and embeddings; Unit I knowledge representation; Unit V evaluation	Competition dataset - see the official page	Add contradiction detection where papers disagree.	DRIVENDATA Research Rovers: AI Research Assistants for NASA Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/252/ai-research-assistants/ Dataset access: Competition dataset (platform account / rules acceptance)
G13	Hriday Khera (12513423) Rakshit Rawal (12510948)	**Public Document Summarisation Without Invented Facts**	Summarising long public documents is exactly where language models invent facts, which is unacceptable for official use.	Government communication teams, journalists, citizens	1. Build a summarisation system that produces a summary plus a factual-consistency score against the source. 2. Produce a summariser with overlap metrics and a separate factual consistency measure. 3. Report rOUGE, factual consistency rate, hallucination count. Baseline: lead-3 extractive summary, mandatory. Deliver Summariser with consistency warnings.	Summarisation; factual consistency checking; evaluation beyond overlap	Unit V NLP; Unit III classification for consistency; Unit V evaluation	Competition dataset - see the official page	Add extractive fallback when the abstractive summary fails consistency.	DRIVENDATA What's Up, Docs? Document Summarization with LLMs Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/297/whats-up-docs/ Dataset access: Competition dataset (platform account / rules acceptance)
G14	Karandeep Singh (12502991) Rahul Sharma (12509643)	**Clinical Note Entity Linking to a Medical Ontology**	Clinical notes are free text, so the conditions they describe cannot be counted, compared or audited without manual coding.	Hospital coding teams, health analytics units	1. Build a linking system that maps clinical mentions to standard ontology concepts with a confidence score. 2. Produce an entity linker with mention detection and linking accuracy reported separately. 3. Report mention F1, linking accuracy, end-to-end F1. Baseline: exact string lookup. Deliver Coder review interface with linked concepts.	Named entity recognition; entity linking to a large ontology; candidate ranking	Unit V NLP and sequence labelling; Unit I knowledge representation; Unit V evaluation	Competition dataset - see the official page	Handle ambiguous mentions by ranking candidates with context.	DRIVENDATA SNOMED CT Entity Linking Challenge Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/258/competition-snomed-ct/ Dataset access: Competition dataset (platform account / rules acceptance)
G15	Bhawandeep (12510219) Yashraj Sharma (12529824)	**Discovering Fall Injury Patterns in Medical Narratives**	Older adult falls are recorded as free-text narratives that nobody reads in aggregate, so prevention is designed on assumption.	Public health agencies, geriatric care programmes	1. Build a discovery system that clusters narratives into recurring circumstance patterns and quantifies each. 2. Produce a clustering of narratives with named themes and their share of incidents. 3. Report silhouette, cluster stability, human-rated theme coherence. Baseline: keyword frequency counts. Deliver Explorer dashboard with theme drill-down.	Text clustering and topic discovery; unsupervised pattern extraction	Unit V NLP and text representation; Unit VI clustering; Unit V cluster validity	Competition dataset - see the official page	Test whether the discovered themes are stable across resampling.	DRIVENDATA Unsupervised Wisdom: Explore Medical Narratives on Older Adult Falls Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/217/cdc-fall-narratives/ Dataset access: Competition dataset (platform account / rules acceptance)
G16	Karanjot Singh Jaswal (12510433) Jobanpreet Singh (12510022)	**Household Poverty Prediction for Programme Targeting**	Poverty targeting relies on surveys that are expensive and quickly outdated, so support reaches the wrong households.	Development agencies, social protection programmes	1. Build a targeting system that predicts household poverty status and reports who is wrongly excluded. 2. Produce a poverty classifier with error rates reported across household subgroups. 3. Report rOC-AUC, exclusion error by subgroup, log loss. Baseline: proxy means test rules. Deliver Targeting dashboard with error breakdown.	Classification on survey data; categorical encoding; fairness-aware error analysis	Unit II encoding and missing values; Unit III classification; Unit V evaluation	Competition dataset - see the official page	Compare inclusion and exclusion error costs explicitly.	DRIVENDATA Power-T Tests: Predicting Poverty Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/50/worldbank-poverty-prediction/ Dataset access: Competition dataset (platform account / rules acceptance)
G17	Hita Khera (12517195) Karanbir Kaur (12505205)	**Automated Classification of School Budget Line Items**	School district budgets are free-text line items that cannot be compared across districts without months of manual coding.	Education departments, school finance analysts	1. Build a classification system that assigns budget lines to standard categories with confidence and escalation. 2. Produce a multi-label classifier with per-label F1 and an escalation rule. 3. Report multi-label log loss, per-label F1, escalation precision. Baseline: TF-IDF logistic regression. Deliver Coding assistant with review queue.	Multi-label text classification; high-cardinality label handling; abstention	Unit V NLP; Unit III classification; Unit V multi-label metrics	Competition dataset - see the official page	Add active learning to reduce the labelling burden.	DRIVENDATA Box-Plots for Education Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/4/box-plots-for-education/ Dataset access: Competition dataset (platform account / rules acceptance)
G18	Ayushi Singh (12512576) Utsav (12506803)	**Industrial Cleaning Process Optimisation**	Industrial cleaning cycles are run to a fixed duration regardless of actual soiling, wasting water, energy and production time.	Process manufacturers, sustainability teams	1. Build a prediction system estimating required cleaning time from process sensor traces so cycles stop when clean. 2. Produce a cleaning-time model with error reported against the fixed-duration baseline. 3. Report mAE, RMSE, resource saved versus fixed cycle. Baseline: fixed duration, mandatory. Deliver Process dashboard with recommended stop point.	Time-series regression on sensor traces; sequence feature engineering	Unit VI time-series; Unit II feature engineering; Unit V regression evaluation	Competition dataset - see the official page	Translate the saving into water and energy per year.	DRIVENDATA Sustainable Industry: Rinse Over Run Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/56/predict-cleaning-time-series/ Dataset access: Competition dataset (platform account / rules acceptance)
G19	Tarnpreet Singh (12513059) Yash Mishra (12516977)	**Building Energy Consumption Forecasting for Demand Planning**	Building operators commit to energy purchases before they know demand, and generic models ignore each building character.	Facility managers, energy retailers	1. Build a per-building forecaster with error reported by building type and time of day. 2. Produce a consumption forecaster beating a seasonal-naive baseline per building. 3. Report mAE, RMSE, error by building type. Baseline: seasonal naive, mandatory. Deliver Energy planning dashboard.	Time-series forecasting; weather and calendar features; per-entity modelling	Unit VI time-series; Unit II feature engineering; Unit V forecast evaluation	Competition dataset - see the official page	Add cold-start handling for buildings with little history.	DRIVENDATA Power Laws: Forecasting Energy Consumption Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/51/electricity-prediction-machine-learning/ Dataset access: Competition dataset (platform account / rules acceptance)
G20	Vrinda (12518126) Kanchan Saini (12505602)	**Anomalous Energy Usage Detection for Fault and Waste**	Faulty equipment and wasted energy hide inside normal-looking consumption totals and are found months later on a bill.	Facility managers, energy auditors	1. Build a detection system that flags abnormal consumption periods without labels and explains the flag. 2. Produce an anomaly detector with precision on manually verified flags. 3. Report precision at top-K flags, detection lead time, false alarms per month. Baseline: fixed percentage-deviation rule. Deliver Facility dashboard with flagged periods.	Unsupervised anomaly detection on consumption series; seasonality handling	Unit VI anomaly detection and time-series; Unit II preprocessing; Unit V rare-event metrics	Competition dataset - see the official page	Separate genuine faults from legitimate operational change.	DRIVENDATA Power Laws: Detecting Anomalies in Usage Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/52/anomaly-detection-electricity/ Dataset access: Competition dataset (platform account / rules acceptance)
G21	Kashish Jain (12517101) Oneal Obrine Garimoto (12501179)	**Cold-Start Energy Forecasting for New Buildings**	A newly metered building has no history, so standard forecasters cannot serve exactly the buildings that most need planning.	Energy service companies, new facility operators	1. Build a forecaster that works from days rather than years of history and reports how accuracy grows with data. 2. Produce a cold-start forecaster with an accuracy-versus-history-length curve. 3. Report mAE by history length, RMSE. Baseline: global average profile, mandatory. Deliver Forecast dashboard for a newly onboarded building.	Transfer and similarity-based forecasting; limited-data modelling	Unit VI time-series; Unit VI clustering for building similarity; Unit V evaluation	Competition dataset - see the official page	Borrow patterns from similar buildings and quantify the benefit.	DRIVENDATA Power Laws: Cold Start Energy Forecasting Organizer: DrivenData (challenge sponsor stated on the official page - VR) https://www.drivendata.org/competitions/55/schneider-cold-start/ Dataset access: Competition dataset (platform account / rules acceptance)

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G22	Hiba Fathima (12512558) Diniya K (12510830)	**Digital Twin Optimisation for Well Production Operations**	Well operating parameters are set from experience and rarely revisited, leaving production and equipment life on the table.	Oil and gas operators, production engineers	1. Build a digital twin that predicts output under candidate settings and recommends a parameter set with expected gain. 2. Produce a surrogate model with prediction error and a recommended setting compared against current practice. 3. Report prediction RMSE, projected gain, constraint violations. Baseline: current fixed settings. Deliver Operator dashboard with recommended parameters.	Regression surrogate modelling; parameter search over the surrogate	Unit III regression; Unit I search and optimisation; Unit V evaluation	Public oil and gas production datasets (Find and cite suitable public data yourselves.)	Add constraints so recommendations respect equipment limits.	SMART INDIA HACKATHON 2026 PS ID: SIH26120 Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field. Organisation: Oil India Limited https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G23	Chirag Khirwar (12504406) Aditya Kashyap (12515990)	**Offset Well Knowledge and Drilling Decision Support**	Drilling decisions repeat mistakes made in nearby wells because the relevant history is buried in unsearchable reports.	Drilling engineers, well planners	1. Build a decision support system that retrieves relevant nearby-well history and surfaces the risks encountered there. 2. Produce a retrieval system with relevance evaluated against engineer judgement. 3. Report recall@K, MAP, extraction F1. Baseline: keyword search over reports. Deliver Well planning interface with nearby-well insights.	Spatial retrieval; document information extraction; similarity ranking	Unit V NLP; Unit I knowledge representation; Unit VI similarity and clustering	Public well log and drilling report datasets (Find and cite suitable public data yourselves.)	Add automatic risk categorisation of retrieved incidents.	SMART INDIA HACKATHON 2026 PS ID: SIH26121 eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations Organisation: Oil India Limited https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G24	Rehan Kashif (12514130) Vasudev Pavithran (12517158)	**Real-Time Airfare Price Index from Web Data**	Official price statistics collect airfares manually from a few outlets, while almost all tickets are now sold online.	National statistical offices, economic policy teams	1. Build an index system that assembles fare observations automatically and produces a transparent price index with coverage reporting. 2. Produce a reproducible fare index with coverage statistics and sensitivity to cleaning choices. 3. Report index stability, coverage, sensitivity to outlier rules. Baseline: simple unweighted mean fare. Deliver Index dashboard with route breakdown.	Data acquisition and cleaning; outlier handling; index construction and smoothing	Unit II data collection, cleaning and outliers; Unit VI time-series; Unit V evaluation	Publicly listed fare data and open travel datasets (Find and cite suitable public data yourselves.)	Add route-level decomposition of index movement.	SMART INDIA HACKATHON 2026 PS ID: SIH26056 Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI). Organisation: MoSPI https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G25	Aastha Kumari (12508826) Ashmit Kumar (12505327)	**On-Device Retail Store Intelligence Platform**	Neighbourhood stores lose sales to stock-outs and queues, and cloud analytics is unusable where connectivity is poor.	Retail store owners, staffing and replenishment managers	1. Build an on-device analytics platform reporting shelf gaps and queue length without sending images anywhere. 2. Produce a detector running on constrained hardware with throughput and accuracy reported together. 3. Report mAP for shelf gaps, queue count MAE, frames per second. Baseline: cloud-scale model for accuracy reference. Deliver Store dashboard with live shelf and queue panel.	On-device object detection; queue estimation; privacy-preserving processing	Unit III-IV CNN and detection; Unit II preprocessing; Unit V detection metrics	Public retail shelf and queue datasets (e.g. SKU-110K) (Find and cite suitable public data yourselves.)	Add stock-out prediction from observed shelf trends.	SMART INDIA HACKATHON 2026 PS ID: SIH26179 To build an AI-powered retail intelligence platform that delivers real-time shopper analytics, automated inventory visibility, and proactive queue management through on-device AI,enabling retailers to reduce stock-outs, improve customer experience, optimize staffing, and increase operational efficiency while maintaining privacy and minimizing cloud dependency. Organisation: Qualcomm Inc https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G26	Ankit Kumar Singh (12516155) Raizel Vodehra (12517123)	**Crowdsourced Societal Problem Matching Platform**	Citizens report problems and student teams look for problems to solve, and the two never meet.	Civic bodies, universities, community organisations	1. Build a platform that clusters submitted problems, removes duplicates, and matches them to teams by skill fit. 2. Produce a deduplication and matching engine evaluated against manual matches. 3. Report deduplication F1, matching precision@K, cluster purity. Baseline: keyword matching. Deliver Civic platform web interface.	Text clustering and deduplication; skill-based matching and ranking	Unit V NLP; Unit VI clustering; Unit V ranking evaluation	Public civic complaint datasets (Find and cite suitable public data yourselves.)	Add a priority score from problem frequency and severity.	SMART INDIA HACKATHON 2026 PS ID: SIH26043 A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships Organisation: Governmcnt of Jharkhand https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G27	Prashant Saxena (12518992) Maraveni Thirupathi (12514991)	**Vernacular Teaching Assistant with Real-Time Translation**	Primary pupils learn best in their mother tongue while material and teachers often are not, and translation quality decides whether it works.	Primary schools, teachers, education departments	1. Build a teaching assistant translating lesson content into the mother tongue with a quality flag on uncertain output. 2. Produce a translation pipeline with quality scores and a flagged-output rate. 3. Report bLEU/chrF against reference, quality-flag precision, human adequacy rating. Baseline: raw MT output, mandatory. Deliver Classroom tool with flagged translations.	Machine translation use and evaluation; quality estimation; abstention	Unit V NLP; Unit III classification for quality estimation; Unit V evaluation	Bhashini and public Indic parallel corpora (Find and cite suitable public data yourselves.)	Add domain glossary enforcement for subject terms.	SMART INDIA HACKATHON 2026 PS ID: SIH26042 AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education Organisation: Governmcnt of Jharkhand https://sih.gov.in/sih2026PS
G28	Gursimran Singh (12505128) Mohamed Faizan (12510441)	**Competency Gap Analysis and Training Recommendation Platform**	Officials are told to upskill from a vast course catalogue with no way to know which courses close their actual gaps.	Training institutions, government HR, learners	1. Build a platform that infers competency gaps from role and assessment data and recommends a ranked learning path. 2. Produce a gap model plus a ranked recommendation list evaluated against expert paths. 3. Report recall@K, NDCG, gap classification F1. Baseline: popularity-ranked course list. Deliver Learner dashboard with recommended path.	Multi-label gap classification; recommendation and ranking; content generation for quizzes	Unit V NLP; Unit III classification; Unit V ranking evaluation	No dataset issued with an SIH problem statement; teams source their own (Find and cite suitable public data yourselves.)	Auto-generate assessment questions from the recommended material.	SMART INDIA HACKATHON 2026 PS ID: SIH26101 Develop an AI enabled learning platform that identifies competency gaps, recommends personalized training through integration with the iGOT Karmayogi ecosystem, and capable of generating Quizzes and Multiple choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System. Organisation: MoSPI https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G29	Shivam Chandel (12522547) Navneet Kumar (12518540)	**Academia-Industry Skill Mapping and Placement Portal**	Students cannot see which skills their target roles require, and employers cannot find candidates who have them.	Universities, students, recruiting employers	1. Build a portal that maps curricula to job requirements and reports each student cohort skill gap against demand. 2. Produce a skill extraction and mapping pipeline with a quantified cohort gap report. 3. Report skill extraction F1, mapping precision, gap coverage. Baseline: keyword skill lists. Deliver Placement dashboard with gap analysis.	Text similarity between curricula and postings; skill extraction; gap quantification	Unit V NLP and embeddings; Unit VI clustering; Unit V evaluation	Public job posting datasets and published curricula (Find and cite suitable public data yourselves.)	Track how demand shifts over time.	SMART INDIA HACKATHON 2026 PS ID: SIH26044 Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement Organisation: Ministry of Ayush https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G30	Ashutosh Kumar Singh (12510696) Davinder Singh (12519413)	**AR Vocational Safety Training Simulator with Performance Scoring**	Industrial safety training is delivered as lectures, so trainees are assessed on recall rather than on what they would do.	Vocational training institutes, mining and manufacturing employers	1. Build a simulator scenario with automatic scoring of trainee actions and a competency report. 2. Produce a scored scenario with scoring agreement against instructor marks. 3. Report scoring agreement with instructors, completion rate, discrimination. Baseline: fixed checklist marking. Deliver Interactive scenario demo with score report.	Rule-based scenario reasoning; action sequence classification; scoring rubric automation	Unit I knowledge representation and rules; Unit III classification; Unit V evaluation	Published industrial safety procedure documents (Find and cite suitable public data yourselves.)	Adapt scenario difficulty to trainee performance.	SMART INDIA HACKATHON 2026 PS ID: SIH26041 AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector Organisation: Governmcnt of Jharkhand https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G31	Yattharth Gupta (12514526) Kandukuri Yaswanth (12514785)	**On-Premise Document Intelligence Workbench for Confidential Data**	Organisations holding confidential documents cannot use cloud AI at all, so they get no document intelligence whatsoever.	Industrial and government document teams, compliance units	1. Build a fully local workbench that answers questions over an internal corpus and never transmits content. 2. Produce a local assistant with answer accuracy and a verified no-network guarantee. 3. Report answer correctness, citation correctness, latency, memory. Baseline: keyword search over the corpus. Deliver Local web workbench with document chat.	Local retrieval-augmented generation; on-premise deployment; grounded answering	Unit V NLP and retrieval; Unit I knowledge representation; Unit V evaluation	Any public document corpus for evaluation (Find and cite suitable public data yourselves.)	Benchmark quality and latency across model sizes on available hardware.	SMART INDIA HACKATHON 2026 PS ID: SIH26117 Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work Organisation: Mangalore Refinery and Petrochemicals Limited (MRPL) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

Group	Students	Project Title	Problem to Solve	Who Benefits	Objectives (what you will be assessed on)	Suggested Techniques	Syllabus Topics Used	Data to Use	Optional Advanced Extension	Inspired By (official source)
G32	Krish Singh (12506651) Sargam (12510800)	**Security Assessment of a Monitoring Web Application**	Monitoring platforms hold sensitive operational data and role-based access that nobody systematically tests before deployment.	Security teams, platform owners, auditors	1. Build a structured security assessment with reproducible test cases and a prioritised remediation report. 2. Produce a documented assessment with categorised findings and reproducible evidence. 3. Report findings by severity, reproducibility rate, false positive rate. Baseline: automated scanner output alone. Deliver Assessment report with reproducible test suite.	Threat modelling; automated test scripting; vulnerability classification and ranking	Unit I problem formulation and search over test space; Unit III classification; Unit V evaluation	A locally deployed open-source monitoring application (Find and cite suitable public data yourselves.)	Automate regression testing so fixes stay fixed.	SMART INDIA HACKATHON 2026 PS ID: SIH26163 Security Assessment of the World Monitor application Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data